Department of Physics, FSK 116, June Supplementary Examination
NEATLY FILL-IN YOUR DETAILS IN THE BOXES BELOW!
Do not write anything else inside the dashed box – not even your signature!

Surname and Names	Student Number

For Official Use ONLY. Apply ✓ if cheat-sheet is compliant (Leave empty if not.
Students who apply ticks here, will lose whatever points they would otherwise earn for their cheat sheet!

Examiners: Mr MJ Legodi
Moderator: Prof WE Meyer

Date: 04 July 2023
Time: 09h30 to 11h00

Total 40

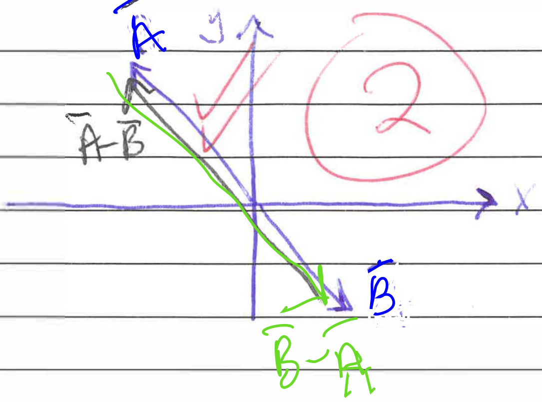
The following specific rules apply to this Test/Examination

1. Fill in your details at the bottom of every page.
2. Answer all questions **ONLY** in the spaces provided. If you need more space, use the "overflow pages" at the back of the question paper and clearly indicate where each question continues. Ask for more overflow pages from invigilators if you require more. No work outside these guidelines will be marked!
3. Pocket calculators may be used, but programmable calculators, cellphones, earphones, electronic watches and other such devices are prohibited.
4. This paper may not be taken out of the test/examination venue. It must be handed in to the chief invigilator, before you leave the venue.
5. Work written in pencil will not be marked. **Write in black permanent pen**.
6. Provide labelled sketches, including co-ordinate systems, free-body diagrams, final answers with correct significant figures and units, etc. Marks will be lost, otherwise.
7. Your answer needs to be written in a systematic and logical fashion. You must show that you understand and can use the relevant Physics. Only substitute numerical values in the final step of calculations! A correct answer does not mean that you will receive full marks.
8. Units must be used when values/numbers are substituted, else marks will be deducted.
9. Extra points will be awarded for high quality or elegant answers.

Constants, conversion factors and some formulae that may be useful
$g = 9.80$ m/s^2 $L_F = 3.33\times10^5$ J/kg $L_V = 2.26\times10^6$ J/kg $c_{water} = 4186$ J/(kg °C) $c_{steam} = 2021$ J/(kg °C) $m_{Earth} = 5.9722\times10^{24}$ kg

QUESTION 1 **[6]**

1.1 Let vector $\vec{A}$ point from the origin into the second quadrant of the xy plane and vector $\vec{B}$ point from the origin into the fourth quadrant. Determine in which quadrant does vector $\vec{B} - \vec{A}$ reside/project (when drawn to point from the origin)? (2)



2

1 pt: Correct magnitudes

1 pt: Correct direction of $\bar{A}-\bar{B}$ or $\bar{B}-\bar{A}$

1.2 Your friend bought a bag of fertilizer for the garden. Its density is given as 100 lb/ft^3. Given that 1 lb = 453.592 g, 1 ft = 12 inch and 1 inch = 25.4 mm; convert the density into SI units. (4)

$100\ lb/ft^3 = ?\ kg/m^3$

$\frac{100\ lb}{ft^3} \times \left(\frac{0.453592\ kg}{1\ lb}\right)\left(\frac{1\ ft}{12\ inch}\right)^3\left(\frac{1\ inch}{0.0254\ m}\right)^3$

$= 1.601845 \times 10^3\ kg/m^3$

$= 1.60\ kg/m^3$

4

(Subtract)

Sig Fig — $\frac{1}{2}$ pt.

Correct value — $\frac{1}{2}$ pt

Incorrect units — $\frac{1}{2}$ pt.

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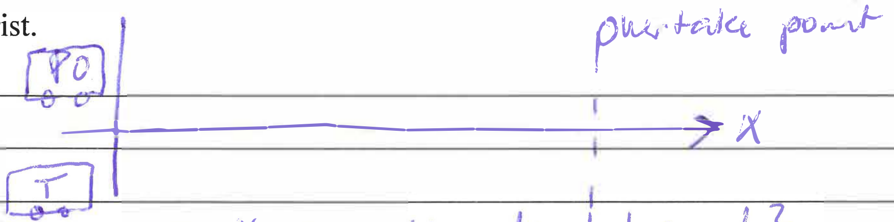
QUESTION 2 **[10]**

A tourist is driving a car along a straight road at a constant speed of 22.5 m/s. Just as she passes a law officer parked along the side of the road, the officer starts to accelerate at 0.50 m/s^2 to overtake her. Assume that the officer maintains this acceleration.

Suppose it is a "slow" day and the officer is a little slow in reacting. Instead of accelerating at the same instant the car passes him, the officer only begins his acceleration 1.00 s after the car passes him.

2.1 From the moment the tourist passes the officer, determine the time it takes the officer to reach the tourist. (5)



$x_{fp} - x_{ip} = v_{xip} t + \frac{1}{2} a_p t^2$ --- (1) {Police Officer

But $v_{xip} = 0$

Also, $x_{f,T} - x_{i,T} = v_{xiT} t + \frac{1}{2} a_x t^2$ --- (2) {Tourist

But $a_{xT} = 0$

The relationship between the time-bases;

$(t_T = t_p + 1.00 s)$ ✓

5

Equating the co-ordinates and simplifying

$\frac{1}{2} a_p t_p^2 = v_{xT,i} t = v_{xT,i}(t_p + 1.00 s)$ when they pass/overtake each other.

$a_p t_p^2 = 2 v_{xT,i} t_p + 2 v_{xT,i}$

$0 = a_p t_p^2 - 2 v_{xT,i} t_p - 2 v_{xT,i}$ ✓

$\therefore t_p = \frac{-(-2 v_{xT,i}) \pm \sqrt{(-2 v_{xT,i})^2 + 4(a_p)(2 v_{xT,i})}}{2(a_p)}$ OR ✓

$= \frac{2(22.5 m/s) \pm \sqrt{(45 m/s)^2 - 4(0.50 m/s^2)(2 \times 22.5 m/s)}}{2(0.5 m/s^2)}$

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$t_p = 45 \pm \sqrt{(45)^2 + 4(22.5)}$ ignoring units

$= 45\,s \pm 45.98912915\,s \Rightarrow t_p = 90.9891\,s$

We reject the negative, unphysical result

2.4 Suppose that the co-ordinates of a particle undergoing motion under constant acceleration in the *xy*-plane are given by: $x_f = x_i + v_{xi}t + (1/2)a_x t^2$ and $y_f = y_i + v_{yi}t + (1/2)a_y t^2$. Show that the vector describing the position of the particle at time *t*, can be expressed as: $\vec{\mathbf{r}}_f = \vec{\mathbf{r}}_i + \vec{\mathbf{v}}_i t + (1/2)\vec{\mathbf{a}}t^2$, where the acceleration, $\vec{\mathbf{a}}$, is a constant. (5)

$\bar{r}_f = x_f \hat{i} + y_f \hat{j}$ with $x_f = x_i + v_{xi} t + \frac{1}{2} a_x t^2$

and $y_f = y_i + v_{yi} t + \frac{1}{2} a_y t^2$

$\therefore \bar{r}_f = (x_i + v_{xi} t + \frac{1}{2} a_x t^2)\hat{i} + (y_i + v_{yi} t + \frac{1}{2} a_y t^2)\hat{j}$

$= (x_i \hat{i} + y_i \hat{j}) + (v_{xi} \hat{i} + v_{yi} \hat{j}) t + \frac{1}{2}(a_x \hat{i} + a_y \hat{j}) t^2$

$= \bar{r}_i + \bar{v}_i t + \frac{1}{2} \bar{a} t^2$

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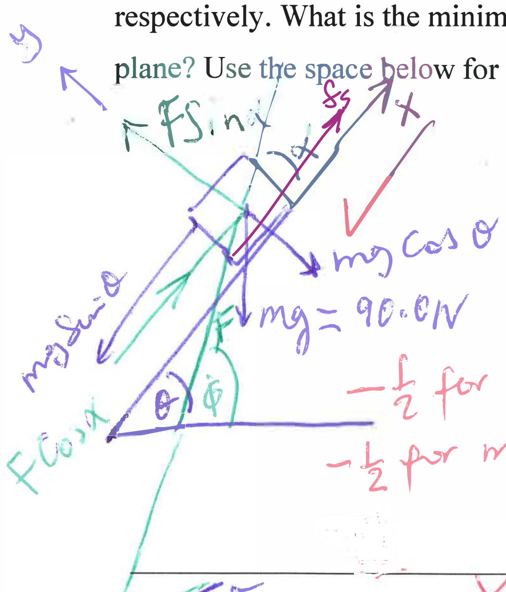
QUESTION 3 **[6]**

3.1 Suppose a block weighing 90.0 N rests on a plane inclined at 25.0° with respect to the horizontal. A force **F** is applied to the object at 40.0° above the same horizontal, pushing it upwards on the plane. The coefficients of static and kinetic friction between the block and the plane are 0.370 and 0.156 respectively. What is the minimum magnitude of **F** that will prevent the block from slipping down the plane? Use the space below for sketches, including free- or force-body diagrams. (6)



$\theta = 25.0°$, $\phi = 40.0°$

$mg = 90.0 N$

Suppose $\alpha = \phi - \theta = 40.0° - 25.0° = 15.0°$

$-\frac{1}{2}$ for missing forces

$-\frac{1}{2}$ for missing axes, directions, angles, etc.

$\sum F_x = ma_x = 0$

$F\cos\alpha - mg\sin\theta + f_s = 0$

$F\cos\alpha - mg\sin\theta + \mu_s F_N = 0$

$\sum F_y = ma_y = 0$

$F_N + F\sin\alpha = mg\cos\theta$

$\therefore F_N = mg\cos\theta - F\sin\alpha$

Substituting F_N into the LHS equation:

$F\cos\alpha - mg\sin\theta + \mu_s(mg\cos\theta - F\sin\alpha) = 0$

$F\cos\alpha - \mu_s F\sin\alpha = mg\sin\theta - \mu_s mg\cos\theta$

$$\Rightarrow F = \frac{mg(\sin\theta - \mu_s\cos\theta)}{(\cos\alpha - \mu_s\sin\alpha)}$$

$$= \frac{(90.0N)(\sin 25.0° - (0.370)\cos 25.0°)}{(\cos 15.0° - 0.370\sin 15.0°)}$$

$= 9.83 N$

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QUESTION 4 (5)

4.1 A wheel rotating about a fixed axis with a constant angular acceleration of 2.0 rad/s^2 turns through 2.4 revolutions during a 2.0 s time interval. What is the angular velocity at the end of this time interval? (5)

$\alpha = 2.0\ rad/s^2$ $\quad \Delta\theta = 2.4\ rev$

$\Delta t = 2.0\ s$

$\omega_f = \omega_i + \alpha t$ We still need to determine ω_i

From $\theta_f - \theta_i = \Delta\theta = \omega_i t + \frac{1}{2}\alpha t^2$ ✓

$\therefore \omega_i = (\Delta\theta - \frac{1}{2}\alpha t^2)/t$ ✓

$= \dfrac{(2.4\ rev)\left(\frac{2\pi\ rad}{1\ rev}\right) - \frac{1}{2}(2.0\ rad/s^2)(2.0\ s)^2}{2.0\ s}$

$= \dfrac{(4.8\pi - 4.0)\ rad}{2.0\ s}$ ✓

(5)

$\Rightarrow \omega_f = \omega_i + \alpha t$ ✓

$= \left[\dfrac{(4.8\pi - 4.0)\ rad}{2.0\ s}\right] + (2.0\ rad/s^2)(2.0\ s)$ ✓

$= 9.5398\ rad/s$

$= 9.5\ rad/s$ ✓

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QUESTION 5 **[13]**

5.1 Consider a particle of mass *m* that is connected to a spring of spring constant, *k*, and its motion described by $x(t) = 10 \sin(\pi t)$.

(a) Utilize the harmonic function given above to write the expressions for the kinetic energy and the potential energy of the particle as functions of time *t*. (2)

(b) At what time (in seconds) is the particle's potential energy equal to the particle's kinetic energy? (3)

$X(t) = 10 \sin(\pi t)$

$K(t) = \frac{1}{2} m v^2 = \frac{1}{2} m \left(\frac{dx}{dt}\right)^2$ with

$\frac{dx}{dt} = [10 \cos(\pi t)] \cdot \pi = 100\pi \cos(\pi t)$

And $\therefore$ $K = \frac{1}{2} m (100)^2 \cos^2(\pi t)$.

$= 50\pi^2 m \cos^2(\pi t)$ ✓ 2

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$U(t) = \frac{1}{2} k x^2 = \frac{1}{2} k (10)^2 \sin^2(\pi t)$

$= 50 k \sin^2(\pi t)$ ✓

From $\omega^2 = \frac{k}{m} \Rightarrow \pi^2 = \frac{k}{m}$ and $m = \frac{k}{\pi^2}$

We have for $K = U$

$50 m \left(\frac{k}{m}\right) \cos^2(\pi t) = 50 k \sin^2(\pi t)$

$50 k \cos^2(\pi t) = 50 k \sin^2(\pi t)$ ✓

$\Rightarrow \tan^2(\pi t) = 1$

$\Rightarrow \pi t = \frac{\pi}{2} = 45^{\circ}$ ✓ 3

$\therefore t = 0.5$ sec ✓

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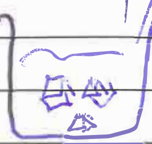
5.2 Steam at 100 °C is added to ice at 0.00 °C. Find the amount of ice melted and the final temperature when the mass of steam is 14.0 g and the mass of ice is 47.0 g. (8)



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$T_{i,S} = 100°C$ $\quad$ $T_{i,ice} = 0.00°C$

14.0 g steam $\quad$ 47.0 g ice

$\Delta Q = 0 \Rightarrow -Q_{hot} = Q_{cold}$ Energy conservation

OR

Determine T_f and how much ice was melted.

$-Q_{hot} = Q_{cold}$

$-(Q_{V,C} + Q_{Hot\ water}) = Q_{F,M} + Q_{melted\ ice\ water}$

$-L_C(m_f - m_i) - m_s c_w (T_{F,Hot\ water} - T_{i,Hot\ water}) =$

$L_{F,m}(m_f - m_i) + m_{ice} c_w T_f - m_{ice} c_w T_{i,ice\ water}$

neg $(L_C m_s) - m_s c_w T_{F,Hot\ water} + m_s c_w T_{i,Hot\ water} =$

pos $L_m m_{ice} + m_{ice} c_w T_f - m_{ice} c_w T_{i,ice\ water}$

Collect terms with T_f and solve for T_f:

$$-T_f = \frac{L_m m_{ice} - L_c m_s - m_s c_w T_{i,Hot\ water}}{-(m_s c_w + m_{ice} c_w)}$$

$$T_f = \frac{L_c m_s + m_s c_w T_{i,Hot\ water} - L_m m_{ice}}{c_w(m_s + m_{ice})}$$

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Substitution .... and calculation of T_f

$T_f = 85.6°C$ water and all the ice has melted i.e. there was sufficient energy from the latent heat process; steam to water to actually melt all of the ice.

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